

AI-Native RAN: 6G Day 1 Operator

Towards AI-Native RAN — China Mobile, arXiv:2507.08403

2026-06-24

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China Mobile(Nan Li, Qi Sun, Chih-Lin I) **operator 6G AI-Native RAN position**
 , “6G release(Day 1) AI native ” 2G~5G .

- 5G AI/ML **add-on** , legacy architecture · protocol · flexibility . 6G AI native .
- **AI for RAN**(AI RAN) **RAN for AI**(RAN pervasive AI compute/connectivity) .

(1) AI-Native RAN essential capability , (2) **AI Node** RAN node Day 1
, (3) 5000+ 5G-A field trial .

(arXiv:2507.08403v1) .

2 Motivation: add-on Day 1

2.1 native

6G RAN AI . (i) massive MIMO signal processing,
(ii) UE · topology radio resource allocation scheduling .
real-time , AI .

2.2 Day 1

release . release 1.5~2 , operator TCO .
4G/5G release(Rel-8, Rel-15) , release vendor , operator .

(**HST**) . HST 2024 48,000 km, 350 km/h , Doppler shift Rel-16
live network . 6G , AI Day 1 “AI-as-afterthought” 6G .

3 essential capability

AI-Native RAN capability (Fig. 1).

3.1 Capability 1 — AI-Driven RAN Processing / Optimization / Automation

L1/L2/L3 AI . AI : , · real-time , .
proactive .

Table I , use case .

- **AI for resources control & optimization (L2/L3):** load balancing, mobility management, network energy saving, network slicing, capacity & coverage optimization, **multi-user scheduling & resource allocation**, QoS/QoE & SLA assurance.
- **AI for air interface (L1):** positioning, beam management, CSI compression/prediction, AI receiver/transceiver, PA non-linearity compensation, interference prediction, RS overhead reduction.

use case inference (BS/UE/LMF-side), model · (DNN/LSTM/DRL/GNN/Transformer, <1M~10M~1B), status(3GPP Rel-18/19, O-RAN, *6G candidate use case*) .

· multi-user scheduling and resources allocation **6G candidate use case** , DNN(<1M), DRL(<10M) . scheduling “AI”
, utility (→ 7).

3.2 Capability 2 — · · AI LCM

· · .

(a) **Data collection.** 5G RAN protocol AI . control plane payload ,
user plane core UPF charging RAN AI , radio link connectivity
:

- *Dedicated AI radio bearer:* RAN · bearer training data · model · metric .
RAN node AI data tunnel .
- *Cross-domain data collection:* application server traffic service QoE RAN .
LLM inference **TTFT(time to first token)** PDU set mapping · DRB configuration .
- *Task-driven customizable collection:* 15 OAM 10 ms latency , UE ,
filter (: UE velocity > 250 km/h) .

(b) **Model management.** off-line mismatch online · fine-tuning pre-validation . monitoring **model performance / network performance / resource utilization** 3 (Table II). UE model context(ID ·) , performance-triggered/periodic retraining, RAN Digital Twin pre-deployment , FL · large foundation model .

(c) **Distributed collaborative computing.** node compute · energy · AI node
· collaborative *training / inference / fine-tuning* , gNB · UE compute capability
logical compute pool resource pooling · dynamic scheduling .

3.3 Capability 3 — AlaaS Provisioning

operator connectivity → edge AI compute → edge AI service enabler . service mode(Fig. 8):

- **Connection assurance for AI services:** multimodal · model update flow QoS framework , **traffic pattern** (embodied intelligence lateral flow, personal assistant *uplink-heavy · bursty* traffic), **deterministic latency**(assistant < 200 ms, embodied AI < 100 ms) .
- **Data services:** traffic load · link quality metric mobility · location sensing . robotic system sensing/update packet loss .
- **RAN AI computing services:** GPU/NPU compute edge AI (- joint scheduling) .

open API framework .

4 Day 1

1. **Day 1** — (HST) .
2. **Forward compatibility** — (hardware) release , centralized resource management + distributed processing site idle compute . (software/ model) AI model processing block · protocol layer .
3. **RAN AI service exposure** — AI compute , idle AI capacity cloud AI connectivity .
4. **3GPP + SDO** — 3GPP end-to-end · interoperability , ITU umbrella framework/ (FG-AINN), ETSI · O-RAN RAN cloud · RAN application(API) . AI Node 3GPP , O-RAN .

5 : AI Node + 6gNB

(Fig. 9). **timeliness** .

RAN OAM	central cloud, non-real-time	supervision · coordination, MDA AI

/		
AI Node	locally centralized, AI compute(edge platform)	data collection, model management, model repository , AI/ML training (L1/L2/L3), L3 inference , RAN capability exposure
6gNB	distributed, UE	(dynamic scheduling) + L1/L2 real-time inference
6G UE		+ LCM/ + () inference.

AI Node 6gNB **one-to-many cardinality** , RAN interface .

:

- **training AI Node** → 6gNB AI compute , sporadic training **statistical multiplexing** (AI BS inference →).
 - **real-time L1/L2 inference 6gNB** . functional split “L1/L2 real-time ”.
 - compute OTT/RAN App AIaaS (new SBI core , OAM).
- timeliness split** “L1/L2 real-time node-local ” . Near-RT RIC xApp externalize node collocate (→ 7).

6 Field Trial

China Mobile 2024 3 trial. AI Node **CCU(Centralized Computing Unit)** .

- **Phase 1** (2024.1~12, **31** · **5000 5G-A**): - AI + use case .
- **Phase 2**: 475,000 gNB , , AI RRM, experience-centric .
- **Phase 3**: , UAV mobility management, embodied intelligence QoE, RAN capability exposure(AIaaS).

6.1 Case 1 —

AI traffic (1000+ app type, 95% ; 8 gNB 4,800) , QoS trigger .
utility :

$$\hat{\mathbf{a}}_u = \arg \max_{\mathbf{a} \in \mathcal{A}} \mathbb{E}[U(\hat{\mathbf{q}}_u(\mathbf{a}, \mathbf{x}_u))]$$

- $\mathbf{a}$: u action(scheduling weight, resource allocation)
- $\mathbf{x}_u$: context(application type, RAN metric, mobility state)
- $\hat{\mathbf{q}}_u(\cdot)$: AI **service-level quality** (latency, throughput, QoE)
- $U(\cdot)$: operator policy utility function(QoE latency)

: poor coverage(RSRP ≤ -105 dBm) · heavy load(PRB $\geq 50\%$) short video latency **33.6%**
/ **30.5%** , **43.0 ms** $\rightarrow$ **32.0 ms (25.6%)**. QR code **18.5 ms** $\rightarrow$ **14.5 ms (21.9%)**.
link 720p $\rightarrow$ 1080p/4K **3~6% traffic** .

6.2 Case 2 — root cause

UE · network · application / :

$$\hat{y}_u = f_{\theta}(\mathbf{x}_u^{\text{App}}, \mathbf{x}_u^{\text{RAN}}, \mathbf{x}_u^{\text{UE}}, \mathbf{x}_u^{\text{Mob}})$$

App (TCP RTT · packet loss), RAN(SINR · RSRP · handover), UE , mobility(grid position-
ing, $50\text{ m} \times 50\text{ m}$) **XGBoost** user type(indoor/outdoor/high-speed) **90% precision**
. rule +20%, recall +30%.

6.3 Case 3 — network energy saving

· · · 20 ES , :

$$\mathbb{E} = \sum_{n=0}^N \sum_{m=0}^M \sum_{t=0}^T \left[P^{PA}(P, m, t) + P^{\text{transceiver}}(m, t) + P^{\text{digital IF}}(m, t, C) \right] + \sum_{n=0}^N \sum_{t=0}^T \left[P^{\text{baseband}}(t, C) + P_0 \right]$$

ES · traffic . AI :

- *Solution 1*: NMS AI 15 cell-level traffic ES $\rightarrow$ **26.6%** .
- *Solution 2*: BS CCU service type , delay-tolerant traffic packet aggregation · sleep $\rightarrow$ **34.16%** .

(AI +5.32%, +12.88%.)

6.4 Trial

CCU multi-vendor gNB cross-platform collaborative inference , AI API
 . → Day 1 (interoperable AI service framework) .

7 gap thesis

“L2 scheduler token importance traffic utility ” . importance-aware MAC scheduling .

()	(thesis gap)
Case 1 utility scheduling ($\arg \max_{\mathbf{a}} \mathbb{E}[U(\hat{\mathbf{q}}_u(\mathbf{a}, \mathbf{x}_u))]$) — operator	$U(\cdot)$ $\hat{\mathbf{q}}_u$ token importance(model entropy · attention) utility → end-to-end task utility/resource PRB/TB granularity
Table I multi-user scheduling 6G candidate use case (DNN/DRL,)	importance-aware L2 scheduling utility · (BLER task utility) (stationary arm/AMR/drone)
Connection assurance: embodied intelligence lateral flow, UL-heavy · bursty traffic, deterministic latency(<100 ms embodied)	mobility offloading · UL scheduling → motivation
Cross-domain data collection: LLM TTFT → PDU set mapping · DRB Task-driven 10 ms data collection, filter(: velocity>250 km/h)	token content predictive resource demand estimation (Stage 3) proactive PRB pre-allocation(chance-constrained, Stage 4)
Timeliness functional split: L1/L2 real-time inference 6gNB(node-local)	“sub-TTI scheduling GPU-collocate Near-RT RIC xApp externalize ”

7.1

- **citation** : China Mobile · Chih-Lin I operator/ → AI-RAN .
- **motivation** : UL-heavy · bursty traffic, robot offloading, deterministic latency use case .
- **gap** : “scheduling utility ” (form) , utility (**token importance**),
 . Stage 3(demand estimation) · Stage 4(proactive pre-allocation) .

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- : *Towards AI-Native RAN: An Operator's Perspective of 6G Day 1 Standardization*
- : Nan Li, Qi Sun, Lehan Wang, Xiaofei Xu, Jinri Huang, Chunhui Liu, Jing Gao, Yuhong Huang, Chih-Lin I (China Mobile)
- arXiv: 2507.08403 (cs.NI), 2025-07-11